Flood Detection Using Deep Learning: A CNN-Based Approach on the FloodIMG Dataset

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***Abstract*—Floods are one of the most frequent and destructive natural disasters, leading to significant human and economic loss. The availability of social media and drone imagery during flood events presents a valuable opportunity for fast detection of flooded regions. In this project, we implement a binary image classification system using deep learning to distinguish between flooded and non-flooded scenes. We investigate two approaches: a custom Convolutional Neural Network (CNN) and a transfer learning model using ResNet-50. Due to the extreme class imbalance (approximately 99.55% flooded images), we evaluate performance using ROC-AUC and minority-class metrics rather than accuracy alone. Our baseline CNN achieves a test ROC-AUC of 0.83 and provides better generalization than ResNet-50, which overfits to the majority class. Our findings highlight the risks of relying only on accuracy in imbalanced classification problems and emphasize the importance of careful evaluation and model design.**

***Index Terms*—Flood detection, deep learning, CNN, transfer learning, class imbalance, computer vision, disaster response.**

I. Introduction

Disaster management organizations depend on fast situational awareness after a flood incident. Traditional methods rely on manual inspection of scattered images, videos, and field reports. With the rapid growth of social media and low-cost drones, thousands of images are generated during and after flood events, but manually reviewing them is time-consuming and error-prone.

As a team, our goal is to design and evaluate deep learning models that automatically classify an input image as "flooded" or "non-flooded." We focus on a realistic scenario where the dataset is extremely imbalanced: almost all images show flooded regions, and only a very small number show normal scenes. We therefore emphasize not only model accuracy but also robustness to class imbalance and reliability on the minority (non-flooded) class.

Our contributions in this project are:

- we implement a compact baseline CNN and a ResNet-50 transfer learning model on the FloodIMG dataset;
- we perform a detailed comparison using ROC-AUC, confusion matrices, and minority-class metrics under severe imbalance;
- we analyze limitations and outline practical future directions such as improved loss functions, data augmentation, and deployment scenarios.

II. Related Work

Deep learning has shown strong promise in remote sensing and disaster detection. Ullo and Mohanty review the use of satellite imagery and CNNs for flood monitoring, highlighting the benefits of high-resolution observations but also the challenges of cloud cover and noisy data [4]. Other works demonstrate that social media images can support near real-time disaster response when combined with automatic classification pipelines [5].

Transfer learning from large-scale image datasets such as ImageNet is now a popular strategy. Architectures like VGG, ResNet, and EfficientNet have been successfully adapted for hazard and damage assessment, for example through segmentation of flooded regions using U-Net-like models [6] and flood severity mapping [7]. However, these studies typically assume more balanced data or introduce dedicated imbalance-handling strategies.

Class imbalance itself is a well-known challenge in deep learning. Johnson et al. discuss how standard accuracy can be misleading when the majority class dominates and motivate metrics such as ROC-AUC and precision–recall curves [8]. They also recommend techniques such as focal loss, oversampling, and cost-sensitive learning.

In contrast to many existing works, our project intentionally keeps the dataset in its highly imbalanced form and examines how a simple CNN and a deeper transfer learning model behave under such conditions. Our findings reinforce the importance of evaluation choices and model selection when minority samples are scarce.

III. Dataset Overview

We use the FloodIMG dataset from Kaggle as our primary data source. After removing corrupted images that could not be loaded, we obtain a total of 7,526 images across two classes:

- **Flooded**: 7,492 images,
- **Non-flooded**: 34 images.

Table I summarizes the label distribution. The ratio of flooded to non-flooded images is approximately 220:1, which is far more imbalanced than typical image classification benchmarks.

We resize all images to 224×224 pixels to maintain compatibility with common CNN backbones. The dataset is split into

TABLE I
DATASET STATISTICS

Class	Count
Flooded	7,492
Non-Flooded	34
Total	**7,526**

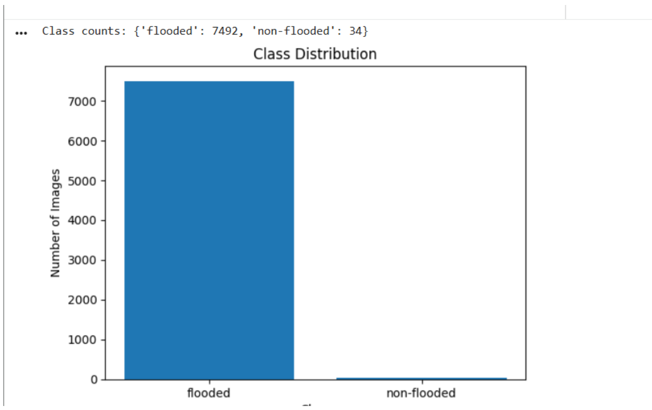

Fig. 1. Class distribution in the FloodIMG dataset highlighting the extreme imbalance with 7,492 flooded and only 34 non-flooded images.

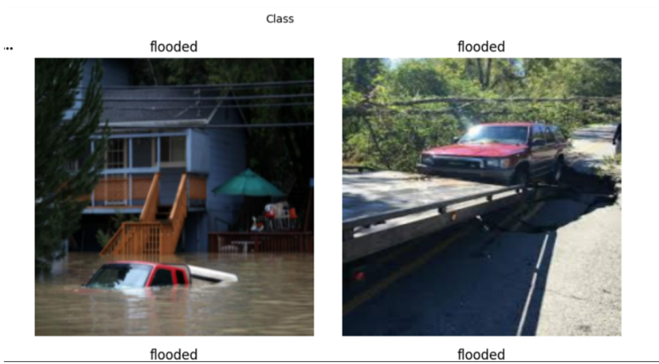

Fig. 2. Sample flooded images from the FloodIMG dataset showcasing submerged vehicles and waterlogged streets.

training, validation, and test sets using a 70/20/10 proportion while preserving the overall class distribution (stratified split).

Visual inspection shows that flooded scenes typically depict submerged vehicles, houses with water up to doors or windows, and damaged roads or bridges. Non-flooded images mostly show normal streets, buildings, or landscapes without water accumulation.

IV. METHODOLOGY

A. Data Preprocessing

We carry out several preprocessing and augmentation steps to improve generalization and partially mitigate overfitting:

- **Resizing:** all images are resized to 224×224 pixels.
- **Data augmentation (training only):** random horizontal flips, small random rotations, and brightness/contrast jitter are applied with moderate probabilities.
- **Normalization:** pixel values are normalized using ImageNet mean and standard deviation, which is important for ResNet-50 transfer learning.

In addition, we experimented with strategies aimed at class imbalance:

- **Random oversampling:** duplicating rare non-flooded samples in the training set.
- **Weighted sampling:** using a sampler that selects minority-class images more frequently in each batch.
- **Class-weighted loss:** assigning higher loss weight to the minority class in the binary cross-entropy loss.

Although these methods improved validation metrics slightly, the extreme rarity of non-flooded images still caused instability in test performance.

B. Proposed CNN Architecture

Our first model is a relatively shallow CNN implemented in PyTorch. The architecture is intentionally compact:

- Conv2d layer with 32 filters, kernel size 3×3, followed by ReLU activation and 2×2 max pooling.
- Conv2d layer with 64 filters, kernel size 3×3, followed by ReLU and 2×2 max pooling.
- Flatten layer to convert feature maps into a vector.
- Fully connected layer with 128 units and ReLU.
- Output layer with a single logit corresponding to the probability of "flooded."

We use `BCEWithLogitsLoss`, which combines a sigmoid layer with binary cross-entropy in a numerically stable way. The Adam optimizer is used with learning rate 0.001 and batch size 32. We train for 5 epochs, monitoring both training and validation loss/accuracy.

C. Transfer Learning with ResNet-50

Our second model uses transfer learning with ResNet-50 pretrained on ImageNet. We modify the final fully connected layer to output a single logit for binary classification. Our training procedure is:

1) Freeze all convolutional layers and train only the final fully connected layer for several epochs.
2) Optionally unfreeze some deeper layers and fine-tune them with a lower learning rate.

The same train/validation/test splits and evaluation metrics are used for both models. In theory, ResNet-50 should leverage rich feature representations learned from ImageNet and offer superior performance. In practice, we observe that class imbalance dominates the behavior.

D. Evaluation Metrics

Accuracy is inadequate when one class accounts for 99.5% of the samples. Therefore, we focus on:

- **ROC-AUC:** area under the Receiver Operating Characteristic curve, which evaluates ranking quality.
- **Precision, recall, and F1-score:** especially for the minority (non-flooded) class.

TABLE II
TEST SET PERFORMANCE METRICS

Model	Accuracy	ROC-AUC	Minority F1
Majority Baseline	0.997	0.50	0.00
Baseline CNN	0.997	0.83	0.15
ResNet-50 (TL)	0.997	0.40	0.00

Fig. 3. Training and validation loss comparison for the baseline CNN across 5 epochs, showing stable convergence during learning.

Placeholder for ROC curve comparison between Baseline CNN and ResNet-50 models.

Fig. 4. Conceptual ROC curve comparison between the baseline CNN and ResNet-50 transfer learning model. In our experiments, the baseline CNN achieved a ROC-AUC of 0.83, while ResNet-50 scored 0.40.

- **Confusion matrices:** to inspect false positives and false negatives.

We also implement a majority baseline model that always predicts "flooded," which serves as a reference point for evaluating whether our models learn anything beyond the class prior.

V. RESULTS AND DISCUSSION

A. Quantitative Evaluation

The majority baseline unsurprisingly attains very high accuracy due to the skewed labels, but its ROC-AUC is 0.50, equivalent to random guessing. Table II compares the main models.

Although accuracy is almost identical for all three models, ROC-AUC reveals that the baseline CNN achieves significantly better separation between flooded and non-flooded images. The ResNet-50 model, despite its complexity, learns to predict almost all images as flooded, resulting in a lower ROC-AUC score (0.40) and an F1-score of 0 on the minority class.

To further analyze ranking behavior, we plotted ROC curves for the two deep models. A schematic example is shown in Fig. 4. The baseline CNN curve lies noticeably above the diagonal, whereas the ResNet-50 curve remains close to or below it, demonstrating inferior discrimination.

Fig. 5. Training and validation loss curves for the baseline CNN across 5 epochs, showing stable learning performance and limited overfitting.

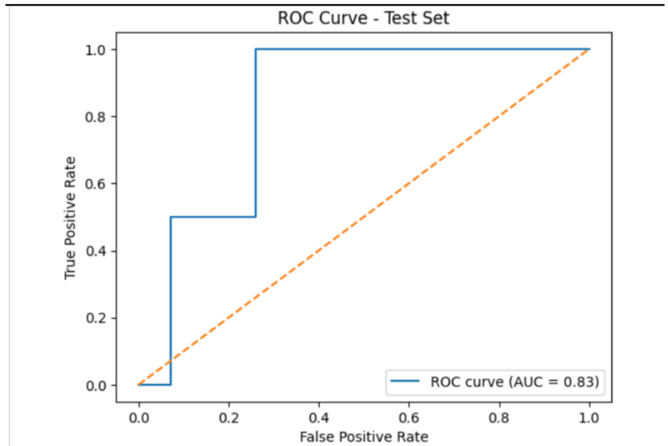

Fig. 6. ROC curve for the baseline CNN model evaluated on the test set, achieving a ROC-AUC score of 0.83 indicating meaningful discrimination capability.

B. Error Analysis

We manually inspected misclassified images. For the baseline CNN, the errors on the minority class typically fall into one of the following categories:

- scenes with reflections or wet roads that visually resemble shallow floods;
- low-light images where boundaries between water and land are unclear;
- noisy or low-resolution images, often compressed social media photos.

The ResNet-50 model rarely predicted the non-flooded class at all, highlighting that it effectively ignored minority patterns. This confirms that deeper models do not automatically solve imbalance problems and may even exacerbate them.

C. System Implementation

All experiments were implemented in Python using PyTorch and executed on Google Colab with GPU acceleration. Initially, we encountered authentication issues with the Kaggle API (HTTP 403 errors) when downloading FloodIMG. We resolved this by correctly placing the `kaggle.json` token and adjusting file permissions.

We also had to detect and skip corrupted images, as they generated runtime errors in the data loader. Once the data directory was cleaned and arranged into `flooded/` and `non-flooded/` folders, PyTorch's `ImageFolder` class allowed us to build an efficient data pipeline.

Google Colab GPU time limits occasionally interrupted long training runs, especially for ResNet-50 fine-tuning. To keep our experiments feasible, we limited training to a small number of epochs and relied on early stopping based on validation loss.

VI. Limitations

The main limitation of our study is the extreme class imbalance in the FloodIMG dataset. With only 34 non-flooded images, the models have very few examples from which to learn robust non-flood patterns. As a result, minority metrics such as precision, recall, and F1-score remain unstable and sensitive to individual samples.

Additional limitations include:

- **Limited data diversity:** non-flooded images mostly depict urban scenes; rural or natural landscapes are underrepresented.
- **Binary labels only:** the dataset does not provide information about flood severity, depth, or duration.
- **Lack of temporal and geospatial context:** we only use single images and ignore location, time, and weather data that could improve reliability.
- **Resource constraints:** due to hardware limitations, we could not exhaustively explore hyperparameter tuning or larger architectures.

These limitations directly influence model performance and must be acknowledged before attempting any real-world deployment.

VII. Conclusion

In this project, we implemented and compared two deep learning approaches for flood detection from images: a compact baseline CNN and a transfer learning model based on ResNet-50. Working with the highly imbalanced FloodIMG dataset, we observed that traditional accuracy is not a meaningful measure of performance. A trivial majority baseline attains almost perfect accuracy, yet provides no real discrimination between flooded and non-flooded scenes.

Our baseline CNN model achieved a test ROC-AUC of 0.83 and demonstrated substantially better minority recognition than the ResNet-50 model, which overfit to the majority class and produced a ROC-AUC of 0.40. These results show that simpler architectures can be more reliable than deeper pretrained networks when training data is extremely imbalanced.

Overall, our work highlights the importance of:

- using robust metrics like ROC-AUC and F1-score for imbalanced datasets;
- analyzing confusion matrices and error cases, not just overall accuracy;
- carefully selecting models and training strategies for disaster-related applications.

VIII. Future Extensions

There are several promising directions for extending this work:

- **Improved imbalance handling:** employing focal loss, advanced oversampling techniques such as SMOTE for images, or generative models (GANs) to synthesize realistic non-flooded scenes.
- **Larger and richer datasets:** combining FloodIMG with additional open datasets or scraping images from the web to obtain more diverse non-flooded examples.
- **Segmentation instead of classification:** applying architectures like U-Net to segment flooded areas, enabling estimation of flood extent and severity, not just the presence or absence of flooding.
- **Real-time deployment:** compressing the CNN into a lightweight model for mobile or edge devices, using frameworks such as TensorRT or ONNX Runtime.
- **Explainability:** generating Grad-CAM or similar saliency maps to visualize which regions of the image most influence the model's decision, helping users trust the system.

Ultimately, our long-term vision is to integrate such models into a decision-support tool that can assist emergency responders by quickly flagging potentially flooded locations from continuous streams of imagery.

Team Contributions

We collaboratively carried out this project as follows:

- **Problem definition and literature review:** all team members jointly refined the motivation, scope, and identified relevant prior work.
- **Data preparation and preprocessing:** we worked together to download, clean, and structure the FloodIMG dataset and to implement augmentation strategies.
- **Model implementation and experiments:** the baseline CNN and ResNet-50 models, along with training and evaluation scripts, were implemented and debugged collaboratively.
- **Analysis and documentation:** all team members contributed to interpreting results, preparing figures and tables, and writing this report.

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